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Oxfordian magnetostratigraphy in the Iberian Range

M. T. Juárez,[1] M. L. Osete,[2] G. Meléndez,[3] and W. Lowrie[1]

Abstract. The magnetostratigraphy has been established in two sections of middle and late Oxfordian age from the Iberian Range (Spain). The polarity sequences correlate well with the magnetozones in sections at Aguilón and Tosos [Juárez et al., 1994]. A composite magnetostratigraphic sequence has been derived for all the sections in the Iberian Range. It correlates with the Handschumacher et al. [1988] extension of the geomagnetic polarity time scale, and ties the oceanic magnetic anomaly sequence to the Oxfordian sedimentary sections.

Introduction

Reversals of the Earth's magnetic field are accurately recorded in the oceanic magnetic anomaly sequence since the Kimmeridgian [Kent and Gradstein, 1985]. However, the polarity time scale remains undefined for most of the earlier Jurassic. The existence of a "Jurassic quiet zone" (JQZ), extending through the Callovian and Oxfordian is controversial [e.g. Kent and Gradstein, 1985, Harland et al., 1990]. There are reports of many reversals of the Earth's magnetic field during the JQZ [e.g. Ogg, 1983, Steiner et al, 1985; Channell et al., 1990; Ogg et al., 1991; Juárez et al., 1994]. Aeromagnetic data also indicate lineated oceanic magnetic anomalies within the JQZ [Handschumacher et al., 1988]. However these reversals cannot be correlated with confidence between the sedimentary and oceanic records.

A previous study of Oxfordian sections in the Iberian Range [Juárez et al., 1994] showed excellent correlation between sections and gave a reliable magnetostratigraphy for this region. We have carried out a detailed study of two additional overlapping sedimentary sections of Oxfordian age, situated in different regions of the Iberian Range. The goal of this study was to correlate these sections with the previously investigated sections in the Iberian Range and with the oceanic record of magnetic anomalies [Handschumacher et al., 1988], and so to correlate and date the reversal sequence in the Oxfordian.

Geological setting and sampling

The Iberian Range, situated in the north-eastern part of the Iberian Peninsula (Fig. 1), has been classified as an intraplate mountain belt extended in a general NW-SE direction and showing a typical aulacogene structure [Alvaro et al., 1979]. Geographically it can be subdivided in two main "branches" of Paleozoic to Mesozoic structural lineations: the Aragonian branch, in the northeast, along the southern margin of the Ebro valley, and the Castillian Branch in the southwest, separated by Tertiary grabens.

The alpine evolution and sedimentary history of the Iberian Basin are determined by the pre-existing Hercynian structures and by the rotation of Iberia during the Cretaceous [Van der Voo, 1993]. The most intense tectonic phase, responsible for the main NW-SE structures of the chain, took place during the Eocene. The sections studied are situated in both branches of this mountain range. The sampled lithologies are shallow water marine limestones of middle and late Oxfordian age. A previous detailed ammonite study carried out in several localities of middle and late Oxfordian ages in the Iberian Range provided a high resolution biostratigraphic scheme involving the recognition of four ammonite biozones, subdivided into 11 subzones [Cariou and Meléndez, 1990; Meléndez and Fontana, 1993]. The new sections also show a detailed ammonite succession that can be correlated well with other Oxfordian sections in the Iberian Range [Meléndez and Fontana, 1993]. The biostratigraphic zonation at the southernmost section investigated (Aguatón, Fig. 1) is rather uncertain due to the scarcity of ammonite fossils.

The two new sections (Fig. 1) are situated near the villages of Aguatón (section AT1: 40.7°N, 1.2°W) and Moneva (section MO1: 41.2°N, 0.8°W). Samples for complementary rock and palaeomagnetic study were also taken from a third outcrop a few kilometers eastward from Moneva (PE1: 41.1°N, 0.7°W), corresponding to the same geological unit as the MO1 section. The attitude (strike/dip) of the bedding is 327°/77.5° NE in the AT1 section and 73°/7°SE at MO1. A total of 140 cores have been sampled and analyzed in both sections.

The investigated sections show a remarkably complete succession of middle and late Oxfordian sediments. The section AT1 has a thickness of 9.3 m while MO1 is 5.25 m thick. Both extend from the Transversarium Zone to the Bimammatum Zone. Thus the mean sedimentation rate of AT1 is double that of MO1. However, the sedimentation rate in the Aguatón

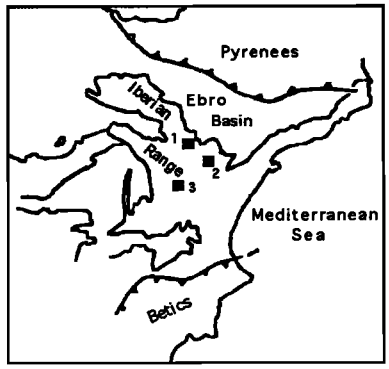


Figure 1. Geographical setting of the investigation area in the Iberian Range (1: Tosos and Aguilón region, 2: Moneva region and 3: Aguatón region).

[1]Institut für Geophysik, ETH-Zürich, Switzerland.
[2]Departamento de Geofísica Universidad Complutense, Madrid, Spain.
[3]Laboratorio de Paleontología, Universidad de Zaragoza, Zaragoza, Spain.

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section is not constant. It is maximum at the lower studied levels of the series (Luciaeformis Subzone) and minimum during the Schilli, Rotoides and Stenocycloides subzones.

A sharp reduction in sedimentation rate is observed at the Transversarium-Bifurcatus Zone boundary in both sections. It involves the presence of a minor hiatus, which partly affects the Rotoides Subzone (top of Transversarium Zone) and the Stenocycloides Subzone (lowermost Bifurcatus Zone). In the Moneva section this is associated with the appearance of a mixed ammonite assemblage formed by reworked fragmentary molds of different ages (top Transversarium to lower Bifurcatus; Fontana, 1990; Meléndez and Fontana, 1991; 1993].

Laboratory procedures

Detailed thermal and AF demagnetisation of the natural remanent magnetisation (NRM) were investigated in pilot samples. On the basis of the results and previous IRM experiments all remaining samples were demagnetized thermally stepwise to 600°C. Heating steps of 100°C were used below 400°C and smaller steps at higher temperature in order to define well the characteristic component of the magnetisation.

Bulk susceptibility was monitored at room temperature after each heating step in order to detect possible mineralogical change during heating. The evolution of this parameter shows the growth of a new mineral phase produced during heating at temperatures higher than 350°C-400°C. This phase carried a strong viscous remanent magnetisation acquired in the laboratory, but it was removed by keeping the samples in field free space for several minutes before each remanence measurement. Palaeomagnetic directions were determined with the aid of vector diagrams [Zijderveld, 1967] and principal component analysis [Kirschvink, 1980].

Description of the results

The magnetic behavior in the new Moneva and Aguatón sections (Fig. 2) is very similar to the Aguilón and Tosos sections [Juárez et al., 1994], with a few minor differences that do not affect the polarity pattern. Two stable components of magnetisation, each different from the present day magnetic field, were isolated in each sample. A low temperature component with a maximum unblocking temperature of 350°C-450°C and a high temperature component with an unblocking temperature of 540°C-580°C were isolated. Following the same nomenclature as in Aguilón and Tosos they are called the S and P components, respectively. Both components are probably carried by magnetite [Juárez et al., 1994], but the different grain-size distributions give distinct spectra of unblocking temperatures. This allows a good estimate of the direction of each magnetisation component.

The magnetisation of the S-component always shows normal polarity while the P-component has alternately normal and reversed polarity. The angle between normal and reversed

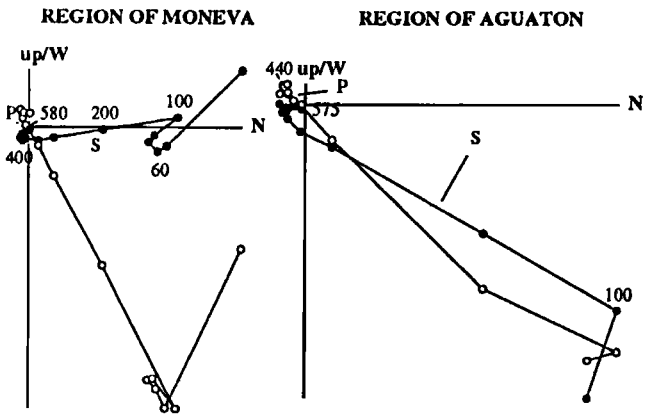


Figure 2. Zijderveld plot for thermal demagnetisation of a typical sample from each site.

directions was 1.5° in Moneva and 3.2° in Aguatón. These compare to critical values of 9.7° and 9.6°, respectively, giving a positive reversal test with classification B [McFadden and McElhinny, 1990].

An incremental fold test performed in earlier studies of the Aguilón and Tosos sections [Juárez et al., 1994; Tauxe and Watson, 1994] showed that the best grouping for both the P- and S-components was achieved after applying 100% bedding tilt correction. Due to their similar magnetic characteristics, the P and S components of magnetisation in the two new sections are considered coeval with the corresponding components in the Tosos and Aguilón sections.

In the section MO1 interpretation is straightforward. The unblocking temperatures of the P and S-components, as well as their palaeomagnetic directions, are equivalent to the values found in the Aguilón and Tosos sections (Table 1). All three sections are inferred to have the same magnetisation history.

In the section AT1 both palaeomagnetic directions are rotated clockwise by approximately 50° with respect to the directions obtained in the other sections (Table 1). However, the rock magnetic properties (unblocking temperature, susceptibility, IRM behavior, etc.) are similar to those found in the Aguilón and Tosos sections. The rotation has been interpreted as a local tectonic rotation of the block containing the outcrop of AT1 [Juárez et al., 1995].

The S-component may constitute a normal (Cretaceous) overprint while the P-component is considered to be the primary component of NRM in all the investigated sections. The polarity changes exhibited by the P-component (Fig. 3) reflect the reversal history of the Earth's magnetic field in the middle and late Oxfordian.

Discussion and conclusions

The correlation between the new sections and the Aguilón and Tosos sections is shown in Figure 4. A refinement of the palaeontological study together with a combination of

Table 1. Palaeomagnetic Directions, Pole Positions, and Statistical Parameters (Fisher, 1953) of P- and S-Components Before and After Bedding Tilt Correction

Region		P-component (no tc)				P-component (tc)						S-component (no tc)				S-component (tc)					
	n	Dec	Inc	k	α_{95}	Dec	Inc	k	α_{95}	Lat	Long	Dec	Inc	k	α_{95}	Dec	Inc	k	α_{95}	Lat	Long
TO/AG	268	309	57	2	9	324	41	10	3	251	56	356	62	2	9	341	45	31	2	234	69
MO/PE	83	332	41	13	5	330	48	12	5	257	63	345	49	52	2	343	56	48	2	258	76
AT	69	286	45	11	5	15	37	11	5	140	68	282	55	47	2	27	34	47	2	124	58

TO/AG: Tosos and Aguilón Region, AT: Aguatón region and MO/PE: Moneva region.

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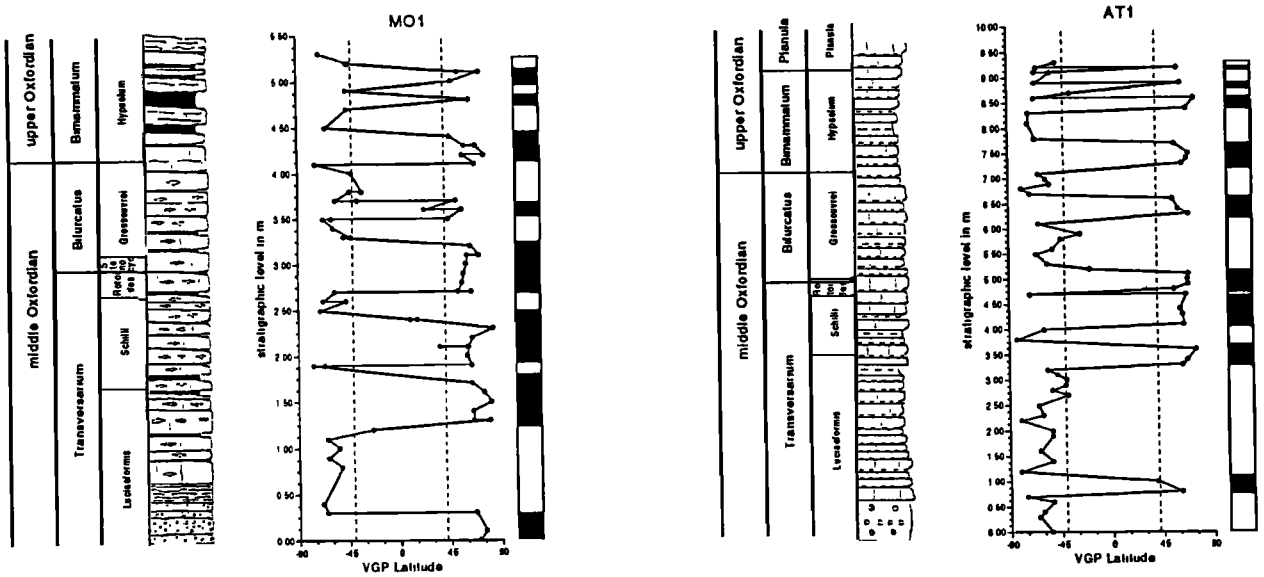


Figure 3. VGP latitude and polarity columns for Moneva (MO1) and Aguatón (AT1).Key to lithostratigraphic symbols: ▭▭▭: limestones, ▬: marls, ooo: iron oolites.

biostratigraphic, lithostratigraphic and magnetostratigraphic data has led to small differences in the placement of the subbiozone boundaries compared to those reported in Juárez et al. [1994]. The minor changes do not affect the magnetostratigraphic correlation of the sections. The position of the Luciaeformis-Schilli Subzone boundary (at about 1.5 m) in the Moneva section is the best compromise between palaeontological and magnetostratigraphic data, although palaeontological data alone suggest a location at about 2.1 m.

The polarity sequences are very similar in all the sections. Single sample magnetozones are present in both new sections. They are considered to be real normal or reversed polarity intervals because they can be correlated with the other studied sections. A composite magnetostratigraphic sequence has been constructed by taking the average thicknesses of the corresponding polarity intervals (Fig. 4). It is characterized by a high reversal frequency during the middle and late Oxfordian.

It can be correlated with other magnetostratigraphic sequences. Ogg and Gutowski [1995] have correlated the Tosos and Aguilón sections to the Polish sections of Ogg et al. [1991]. The polarity pattern for the Transversarium and Bifurcatus Zones in our composite sequence for the Iberian Range agrees well with the Picola Mantova II section in northern Italy [Channell et al., 1990].

In addition the composite sequence can be correlated with the oceanic magnetic anomaly record derived by Handschumacher [1988] (Fig. 5). The beginning of our sections (lower part of Transversarium Zone) corresponds to anomaly M30a (Handschumacher nomenclature), which is correlated with magnetozone R1 in our composite sequence. R2 corresponds to oceanic magnetic anomaly M30 . This reversed polarity interval is followed in both oceanic and composite sequences by a long normal polarity interval. It contains three reversed intervals in the oceanic sequence (M29, M28b and M28a) and two in our composite sequence (R3 and R4). A tentative explanation is that the short reversal appearing within the N2 interval in the section AG2 is a real one which would correspond to the oceanic chron M29. On the other hand, the hiatus in the Transversarium-Bifurcatus Zone boundary is situated at the upper part of this normal polarity period. The lack of sedimentation at this level would have an effect on all the studied sections in the Iberian Range and may have been underestimated in the Aguilón and Tosos sections [Juárez et al., 1994].

Later in the series R5, R6 and R7 are correlated with M28, M27 and M26, respectively. After this period a very high frequency of reversals occurs in both our composite sequence and the M-sequence.

Our study has four important consequences. First, it slightly affects the geomagnetic polarity time scale of Gradstein et al. [1994]. These authors considered two different polarity patterns for the Callovian and Oxfordian stages: the magnetostratigraphic polarity sequence based on the work by Steiner et al. [1985] in the Iberian Range and Ogg et al. [1991] in south Poland, and the extension of the oceanic record by Handschumacher et al. [1988]. Our data allow us to determine a unique magnetic pattern for the middle and late Oxfordian stages. The main difference between the Gradstein et al. [1994] correlation and ours lies in the precise location of the biozone

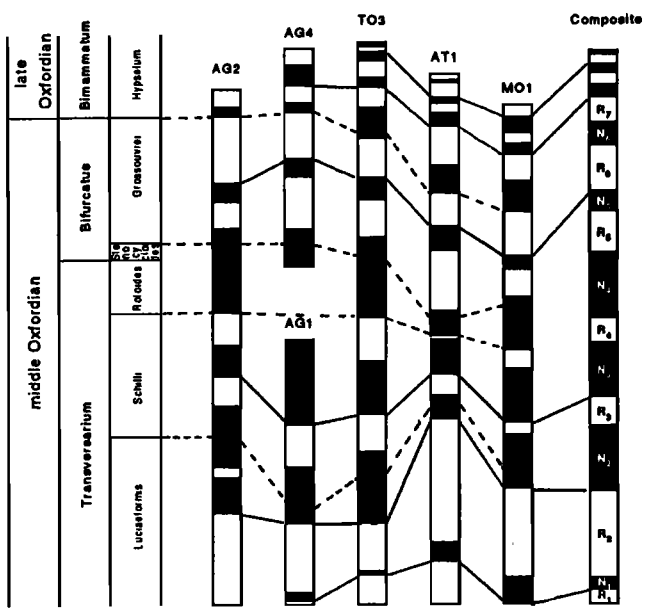


Figure 4. Correlation of all sections in the Iberian Range (Tosos and Aguilón sections from Juárez et al., 1994) and the resulting composite sequence.

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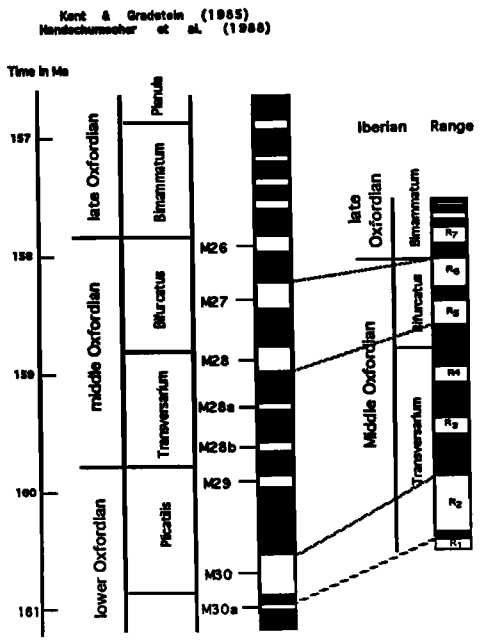


Figure 5. Comparison between the composite sequence and the oceanic record extended by Hnadschumacher [1988]. The time scale and biozonation are from Kent and Gradstein [1985].

boundaries with respect to the polarity chrons. Our correlation between magnetic polarity intervals in the sedimentary and marine magnetic records agrees with that proposed by Ogg and Gutowski [1995], and adds new data that better constrain the geomagnetic time scale.

Second, assuming a uniform rate of sea-floor spreading during the middle and late Oxfordian, our proposed correlation contradicts the hypothesis of equal duration of ammonite zones commonly assumed in the derivation of time scales. Our data suggest that the Transversarium Zone lasted twice as long as the Bifurcatus Zone.

Third, our results clarify the uncertain origin of some low amplitude, high frequency anomalies in the oceanic record [Cande and Kent, 1992]. The early M-sequence anomalies indeed represent geomagnetic reversals.

Finally, our proposed correlation tentatively resolves some uncertainties in the identification and correlation of the Oxfordian polarity pattern of the Earth's magnetic field and provides new information for the definition of a geomagnetic polarity time scale for the Jurassic.

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M. T. Juárez, Institut für Geophysik, ETH-Hönggerberg, CH-8093 Zürich, Switzerland. (e-mail: teresa@mag.geo.phys.ethz.ch)

M. L. Osete, Departamento de Geofísica Facultad de Física, Universidad Complutense, 28040 Madrid, Spain.

G. Meléndez, Laboratorio de Paleontología. Departamento de Geología, Facultad de Ciencias, Universidad de Zaragoza, 50009 Zaragoza, Spain.

W. Lowrie, Institut für Geophysik, ETH-Hönggerberg, CH-8093 Zürich, Switzerland.

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